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EDUCATION

New Mexico State University

Las Cruces, NM, USA

Master of Science in Industrial Engineering (In Progress)

Aug 2025 – Present

- **Relevant Coursework:** Advanced Production and Inventory Control, Database Management Systems, and Statistical Methods for Aluminum Manufacturing.

National University of Science and Technology

Islamabad, Pakistan

Master of Science in Mathematics

Sept 2022 – July 2024

- **CGPA:** 3.90/4.00
- **Thesis:** Algebraic Invariants associated with Bristled Cubic Circulant Graphs.
- **Description:** Extensively explored the algebraic invariants of the whisker cubic circulant graphs. Considered the associated edge ideal in the polynomial ring Q over a field F and investigated its fundamental characteristics, including regularity, depth, Stanley depth, and projective dimension.

International Islamic University

Islamabad, Pakistan

Bachelor of Science in Mathematics

Sept 2018 – Jan 2022

- **CGPA:** 3.65/4.00

PUBLICATIONS

- **Afridi, M. U. K., Sohn, H., & Kwakye, J. (2026).** Improving spare parts procurement decisions under limited budgets: A predictive-optimization framework with a criticality-threshold property. *Manuscript submitted for journal review.*
- **Afridi, M. U. K. (2026).** GEOSUPPLY: A geopolitical oracle-mediated governance framework for blockchain supply chain smart contracts under force majeure. *SSRN Working Paper.* ssrn.com/abstract=6580131
- **Afridi, M. U. K. (2026).** BCRRS: A blockchain-anchored contractor reputation system with ML ranking, anti-nepotism governance, and cross-border intelligence for construction procurement. *SSRN Working Paper.* ssrn.com/abstract=6607858
- **Afridi, M. U. K., Rehman, I. U., & Ishaq, M. (2025).** Algebraic invariants of the edge ideals of whisker graphs of cubic circulant graphs. *Heliyon*, e41783. [doi:10.1016/j.heliyon.2025.e41783](https://doi.org/10.1016/j.heliyon.2025.e41783)
- **Ur Rehman, I., Afridi, M. U. K., Ishaq, M., Asiri, A., & Hussain, A. (2025).** Algebraic invariants of edge ideals of some bristled circulant graphs. *AIMS Mathematics*, 10(5): 11330–11348. [doi:10.3934/math.2025515](https://doi.org/10.3934/math.2025515)

PROFESSIONAL EXPERIENCE

New Mexico State University

Las Cruces, NM

Graduate Teaching Assistant – Engineering & Operations Courses

Aug 2025 – Present

- **Courses Supported:** Engineering Economy, System Thinking and Decision Making, Linear Programming, and Ordinary Differential Equations (ODE).
- **Academic Oversight:**
Delivered structured office hours for Linear Programming and Engineering Economy, translating complex optimization concepts into applied industrial contexts

Stanford University Code in Place 2025 & 2026

Stanford University

Section Leader – Python Instructor

April 2025 – June 2025 & April 2026 – Present

- Selected as a Section Leader for two consecutive years (2025 and 2026) for Stanford University's Code in Place program, a global Python programming course open to learners from over 100 countries.
- Taught weekly Python sections to a cohort of 10–15 students, covering programming fundamentals, algorithmic thinking, and problem-solving through structured lessons and live Q&A sessions.

- Provided individual feedback on student code and supported learners across different time zones and educational backgrounds. ([View Certificate](#))

TECHNICAL SKILLS

- **Programming & Data Analysis:**
 - Python (Pandas, NumPy, Scikit-learn, XGBoost, Prophet); R; SQL
- **Forecasting & Time Series:**
 - ARIMA, SARIMA, Holt's Method, Double Exponential Smoothing, Moving Averages — implemented in both Python and Excel/Minitab; benchmarked against ML methods (XGBoost, Random Forest, Facebook Prophet) on Walmart retail dataset
- **Optimization:**
 - Linear Programming, Predict-then-Optimize framework (published); LINDO/LINGO (sensitivity analysis); Python (SciPy, MILP)
- **Statistical Quality Control:**
 - R (control charts, UCL/LCL, outlier detection, Type I & II error analysis); Minitab; applied in Statistical Analysis in Manufacturing coursework
- **Databases & Tools:**
 - SQL (joins, subqueries, stored procedures, normalization); MongoDB; CockroachDB; Excel (Solver, pivot tables); $\text{\LaTeX}$; MATLAB

SELECTED RESEARCH PROJECTS

- **ProactiveCPM (Live-Signal RAG + Reinforcement Learning for Construction Scheduling):**
 - Designed a six-component framework combining a project-specific retrieval layer (LiveRAG) over five live operational signal channels (IoT sensors, supplier communications, HR systems, weather APIs) with a GNN+PPO reinforcement-learning agent to proactively manage Critical Path Method (CPM) schedules; evaluated on 1,500 simulated construction projects, achieving a 20.6% mean delay reduction versus static CPM ($p < 0.001$), preventing 38.6% of catastrophic overruns, and generating LLM-based project-manager briefings in under 30 seconds versus the industry-standard 24-hour reactive lag.
- **DentaAgent (Judge-Model Agentic RAG):**
 - Engineered a clinical decision support system utilizing a 70B LLM to synthesize SDCEP guidelines and drug interaction data. Implemented a bidirectional human-in-the-loop pipeline that triggers automated model retraining based on expert clinical feedback.
- **SensorGuard (Multi-Agent AI):**
 - Developed a seven-specialist agent framework for real-time carbon dioxide fraud detection. Integrated stoichiometric ratio validation and Byzantine spread detection to identify coordinated multi-sensor conspiracies, achieving an F_1 -score of 0.957.
- **Demand Forecasting Benchmarking Study (Walmart Dataset):**
 - Compared classical time series methods (ARIMA, SARIMA, Holt's, double exponential smoothing) against modern ML approaches (XGBoost, Random Forest, Facebook Prophet) to identify scenario-specific performance gaps; implemented in Python and Excel/Minitab.
- **Statistical Process Dashboard (R/Shiny):**
 - Built an interactive R Shiny dashboard for the Statistical Methods for Aluminum Manufacturing course, translating control charts, hypothesis testing, and process capability analysis into an accessible, interactive visualization tool for manufacturing quality decisions. ([View Dashboard](#))

PRESENTATIONS

NMSU Graduate Research Symposium — Spring 2026

New Mexico State University, Las Cruces, NM

Poster Presentation

Spring 2026

- Co-presented research on an IoT-enabled predictive digital twin framework for real-time monitoring and decision support in smart additive manufacturing, with co-author Calvin Cox.
- Supervised by Dr. Janet Kwakye and Dr. Hansuk Sohn, Department of Industrial Engineering, NMSU.

NUST Conference on Recent Trends in Mathematical Sciences-IV (RTMS-IV)

NUST University, Islamabad,

Pakistan

- Presented research paper: *Depth, Stanley depth, and the regularity of the edge ideal of bristled cubic circulant graphs* at an international mathematics conference. ([View Certificate](#))

International Poster Competition, Mathematics Media

Online

Poster Presentation – 2nd Place

August 14, 2024

- Presented on cryptographic methods including chaotic maps and modern encryption schemes; awarded 2nd place among 22 teams from NUST University. ([View Certificate](#))

PEER REVIEW & EXPERT JUDGING**Engineering Applications of Artificial Intelligence, Elsevier**

International

Peer Reviewer

2025 – Present

- Invited to peer review manuscripts submitted to this Elsevier journal (Impact Factor ~8.0, Scopus and Web of Science indexed) covering AI and engineering applications.
- Evaluated methodological soundness, technical accuracy, and contribution to the field; provided structured feedback to authors and the editorial board.

New Mexico Technology Student Association State Conference

Las Cruces Convention Center, NM

Judge, Data Science & Analytics Competition

2026

- Served as a technical judge at the NMTSA 2026 State Conference, attended by over 400 students from 26 chapters across New Mexico.
- Evaluated student projects on analytical methodology, data interpretation, and practical application; scored and ranked competing teams.

NMSU Spring Undergraduate Research Symposium

New Mexico State University, Las Cruces, NM

Research Poster Judge

Spring 2026

- Invited by NMSU to evaluate undergraduate research posters across disciplines, assessing research quality, methodology, and clarity of presentation.

Southern New Mexico Science Olympiad Regional Tournament

Las Cruces, NM

Event Judge, Helicopter Design

Feb 2026

- Evaluated student-engineered helicopter designs for flight duration and structural stability; provided technical feedback to participants.